Zahra Golpayegani

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About Me

I am a machine learning engineer with **4+ years of experience scaling and optimizing** large language models for **efficient training and deployment**. My work focuses on **hardware-aware** performance optimization, p**arallel and distributed systems** design, and **end-to-end ML infrastructure** across training and inference stacks.

Experience

Huawei Canada – *Researcher* Sep 2024 – Present
Tools: PyTorch, vLLM, VeRL, Megatron-LM, DeepSpeed, HuggingFace, LLaMA Factory

- Integrated **async RL training with multi-turn tool-call support for Ascend NPUs** in **VeRL** (CANN-recipes-train repository), enhancing long-horizon reasoning performance and extending **evaluation** coverage with benchmarks such as LiveCodeBench (PR #101, PR #102).
- Implemented a **block-wise speculative decoding method** (proposed by Google) and integrated it into **vLLM and vLLM-Ascend**, improving inference **throughput by up to 10.2%** and reducing **TPOT by 9.2%** (PR #40819, PR #7808).
- Co-developed a simulation library for **distributed LLM training and serving workloads**, incorporating kernel-level FLOP and memory-based **performance modeling** for new LLM architectures, enabling **systematic throughput and latency analysis** across hardware and serving configurations.
- Implemented an **Attention-FFN Disaggregation (AFD)** simulation framework for SOTA LLMs (e.g., **DeepSeek NSA, DeepSeek DSA, TurboS**) to evaluate serving efficiency in disaggregated pipelines.
- **Profiled** and identified **performance bottlenecks** in LLM workloads on **Huawei Ascend chips**, proposing **optimization** strategies to improve compute efficiency.
- Benchmarked parallelization strategies—including **tensor, expert, context, sequence, and data parallelism**—for models such as **DeepSeek-NSA** and **MiniMax-Text-01**, quantifying impacts on memory usage, communication overhead, and throughput.
- Accelerated pre-training of **LLaMA-based models** using conditional computation, reducing training cost by **up to 16%** and improving MFU by 3% without significant accuracy loss.
- Co-authored a paper (ETT) proposing a **test-time training** method that extends GPT-Large and Phi-2 **context length by up to 32×** while **improving accuracy by 30%**.

Zetane Systems – *AI Engineer* Dec 2021 – Aug 2024
Tools: Python, LangChain, OpenAI API, PyTorch, Docker, Git, MLflow

- Co-designed and developed ZetaForge, an AI platform automating prompt-to-pipeline generation using LLMs.
- Developed complex end-to-end multi-agent AI pipelines leveraging state-of-the-art libraries such as **LangChain** and **openai-python** (blog post).
- Implemented a robust **MLOps pipeline** (end-to-end machine learning operations workflow) featuring data and model validation, preprocessing, model training with **automated experiment tracking**, and continuous model improvement (open-source project).

- Contributed to flagship products including Protector—a platform for assessing vision model reliability in real-world conditions.
- Delivered product demos and technical talks to prospective clients and international audiences at events including Scale AI, World Summit AI, and ALL IN.

XAI Lab, Concordia University – *Research Assistant* Sep 2021 – Jan 2024
Tools: Python, PyTorch, OpenCV, MATLAB, LaTeX

- Investigated the interplay between **model robustness, accuracy, and shape bias** under Professor Nizar Bouguila's supervision; published findings at **CRV 2023**.
- Proposed a novel image compression algorithm inspired by **Singular Value Decomposition** (SVD) to optimize robustness of vision models trained on compressed data; work published at **ICPRAM 2024**.

Skills

Programming Languages: Python, C/C++
MLOps & DevOps: Docker, LangChain, MLflow, Git
Deep Learning Frameworks: PyTorch, vLLM, VeRL, Megatron-LM, DeepSpeed, HuggingFace, LLaMA Factory

Education

Concordia University, MASc Information Systems Engineering 2024
GPA: 4.0 / 4.0
Thesis: Enhancing Deep Learning Model Robustness: Insights from Out of Distribution Data Augmentation and an Innovative Image Compression Technique
Supervisor: Prof. Nizar Bouguila
Amirkabir University of Technology, BSc Computer Science 2020
GPA: 3.7 / 4.0
Final Project: Food9K: Detecting Food in Social Media Images Using YOLOv3
Supervisor: Prof. Mohammad Akbari

Publications

- ETT: Expanding the Long Context Understanding Capability of LLMs at Test-Time. 2025. arXiv.
- PatchSVD: A Non-Uniform SVD-Based Image Compression Algorithm. 2024. ICPRAM.
- Clarifying myths about the relationship between shape bias, accuracy, and robustness. 2023. CRV.

Awards

- **Campaign Hero Award**, Huawei *2025*
  Honored for delivering rapid, high-impact results on a key internal campaign at Huawei HQ.
- **Future Star Award**, Huawei *2025*
  Recognized for outstanding performance and potential, highlighting my contributions.
- **Best Paper Award Candidate**, ICPRAM *2024*
  Nominated for research on model compression and robustness.
- **Conference and Exposition Award**, Concordia University *2023*
  Awarded for excellence in presenting graduate-level research.
- **Accelerate Explore Award**, Mitacs *2021*
  For conducting applied AI research in collaboration with industry partners.
- **Merit Scholarship Entrance Award**, Concordia University *2021*
  Awarded for high academic achievement upon admission to MASc program.
- **2nd Place, Worldwide**, RoboCup Junior Soccer League *2013*
  RoboCup World Competitions – international robotics tournament.